

Online Removal of Cardiac Artifacts from Respiratory Surface Electromyograms

J. Petrick * J. Sauer * J. Graßhoff * P. Rostalski *

** Institute for Electrical Engineering in Medicine, Universität zu Lübeck, 23562 Lübeck, Germany (e-mail: josefine.petrick@outlook.com, j.sauer@uni-luebeck.de, jan.grasshoff@uni-luebeck.de, philipp.rotalski@uni-luebeck.de).*

Abstract: Surface electromyographic (sEMG) signals of the respiratory muscles can be used for assessing patient activity during mechanical ventilation. However, the measured signals are usually heavily contaminated by the electrical activity of the heart. In this paper, an online algorithm for real-time respiratory sEMG denoising by means of the stationary wavelet transform (SWT) is presented. It was analyzed on a dataset of healthy patients and was found to perform comparably to an offline SWT algorithm on several metrics. Online processing offers the potential to improve the therapy of ventilated patients directly at the bedside.

Copyright © 2024 The Authors. This is an open access article under the CC BY-NC-ND license (<https://creativecommons.org/licenses/by-nc-nd/4.0/>)

Keywords: respiratory sEMG, online algorithm, signal processing, mechanic ventilation, removal of cardiac artifacts

1. INTRODUCTION

During mechanical ventilation, it is essential to monitor the patient's spontaneous breathing. This allows the observation of the respiratory effort and the temporal course of respiratory activity. Just as too high or too low breathing effort can be harmful to the patient, also high patient-ventilator asynchrony can cause severe injuries (Heunks and Ottenheijm, 2018). Therefore, straightforward approaches for continuous quantification of respiratory activity at the bedside are needed (Mauri et al., 2017). A reliable inspiration detection could also serve as a trigger for a mechanical ventilator. This would have significant clinical implications, as conventional triggers can be difficult to adjust and require the patient to have sufficient strength to activate them (Larsen and Ziegenfuß, 2009). Additionally, patient-ventilator asynchronies affect approximately 25 % of mechanically ventilated patients (Bakkes et al., 2023). The detection of inspiration using a surface electromyographic (sEMG) signal might be one possible solution (Sauer et al., 2024). An advantage of the sEMG in comparison to other existing measuring techniques, such as the ones based on nasogastric catheters, which measure for example Pes, is that it is a non-invasive and straightforward method for retrieving information about the activity of the diaphragm (Rafols-de Urquía et al., 2018). However, as respiratory sEMG measurements often are heavily contaminated by electrocardiographic (ECG) artifacts, they need to be filtered in order to achieve viable respiratory signals (Petersen et al., 2020).

Currently, a wide range of offline algorithms for the purpose of ECG removal in respiratory sEMG measurements exists. These include template subtraction, model-based filtering, empirical mode decomposition (EMD), simple high-pass filtering, and stationary wavelet transforms (SWT) (Petersen et al., 2020). From these algorithms, the simple highpass, the EMD, and the SWT

were most effective in suppressing ECG artifacts, refer to Petersen et al. (2020) for more details. However, in order to use sEMG signals for real-time monitoring or as a trigger for mechanical ventilation, an online denoising algorithm is essential. Even though highpass filtering could easily be implemented online, the results are often not satisfactory as it heavily distorts the EMG's spectrum. It is therefore advisable to use more complex algorithms such as EMD or SWT for filtering the raw sEMG signals. Additionally, other algorithms for the real-time removal of ECG parts from sEMG signals have been published as the one from Hou et al. (2022). However, contrary to the proposed algorithm, their algorithm has not yet been tested online.

In the following, an online implementation for single channel respiratory sEMG measurements is presented. It is derived from an offline algorithm published by Petersen et al. (2020) and uses a stationary wavelet transform to remove most of the ECG while preserving the main parts of the sEMG. Other sources of interference of the signal are not considered. The algorithm relies on an adaptive thresholding scheme, which requires QRS detection to be performed beforehand. Several well-established online QRS detection algorithms exist such as the one of Pan and Tompkins (1985) or more recently from Christov (2004). In this paper, the results of the proposed online ECG removal algorithm are analyzed and verified using real respiratory sEMG signals of six subjects and synthetically generated signals based on compositions of a clean sEMG and a clean ECG signal at different SNRs.

2. METHODS

2.1 Algorithm

The proposed online algorithm is able to process single-channel respiratory sEMG measurements. Usually, the

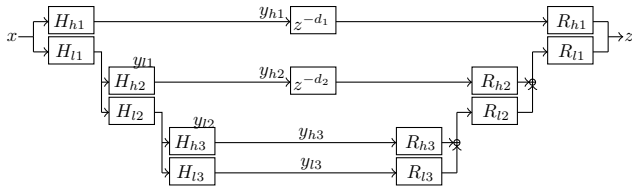


Fig. 1. The structure of a three-layer filter bank with input x , output z , highpass-filters H_{hi} and R_{hi} and lowpass-filters H_{li} and R_{li} , $i \in \{1, 2, 3\}$ defining the level. While the H filters are for decomposition, the R filters reconstruct the signal.

sEMG should be sampled with at least 512 Hz, but in many cases, 1024 to 2048 Hz will give the best results. A python implementation of the proposed algorithm is made available on GitHub¹.

In order to efficiently remove cardiac artifacts, the algorithm uses the stationary wavelet transform (SWT) with a Daubechies wavelet (Daubechies and Sweldens, 1998). Wavelet transforms are widely used in biomedical signal processing. They decompose a signal into multiple frequency bands, thus, enabling frequency-dependent filtering. Since they can be implemented as a filter bank, they are nicely suited for online implementation. For 1024 Hz we recommend using at least three to four wavelet levels. A good rule of thumb is to add one additional layer when doubling the sampling rate. For the validation in this paper, a three-layer filter bank with the db2 wavelet and a sampling rate of 1024 Hz was used.

Wavelet transforms are especially useful for sEMG-ECG separation as an ECG signal typically contains lower frequencies than an sEMG signal. However, the QRS complexes of an ECG signal contain particularly high frequencies that overlap with those of the sEMG signal. Therefore, the segments affected by QRS complexes have to be detected and treated separately. To this end, the online algorithm applies an adaptive threshold to each wavelet band. Within the QRS regions thresholds are more sensitive to remove as much as possible from the peaks. Between QRS regions thresholds are less sensitive. The frequencies in each wavelet band determine the number of samples to which the more sensitive threshold is applied. As the lower-frequency wavelet bands are believed to contain more ECG, the more sensitive threshold is applied here to larger parts. A flow diagram of the online algorithm is represented in Figure 3.

In order to locate the QRS regions during real-time processing, an online QRS detection is needed. However, the peak detection has softer real-time requirements than the actual filter algorithm because the position of the next peak can be predicted as we will see later. In this paper, a QRS detection based on the algorithm of Pan and Tompkins (1985) with a delay of 300 ms is used. Since the ECG removal is a causal filtering algorithm, the QRS detection is also implemented by means of causal filters. Other online capable QRS detectors can be considered as alternatives to Pan and Tompkins. The remainder of the paper focuses on the online ECG removal by means of an SWT.

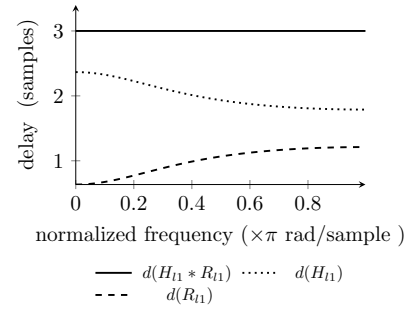


Fig. 2. The behaviour of the group delays in dependence of frequency exemplary for level 1 of a filter bank using the db2 wavelet. Refer to Figure 1 for definition of the wavelet filters H_{i1} and R_{i1} .

Structure of the filter bank implementation The stationary wavelet transform is implemented as a filter bank consisting of an equal amount of decomposition and reconstruction filters referred to as H_{wi} and R_{wi} respectively, where i corresponds to the level of the filter. While $w = h$ indicates a high-pass filter, an l indicates a low-pass filter. These FIR filters are also implemented online as causal filters. They are connected as shown in Figure 1 for a three-layer filter bank. While the SWT corresponds to the decomposition part of the filter bank, the inverse SWT (ISWT) corresponds to the reconstruction. Between decomposition and reconstruction, the detail coefficients y_{hi} are truncated by means of an adaptive thresholding method dependent on the layer as explained above. Since the delay induced by the filters increases with each level of decomposition, the delays of the higher levels z^{-d_i} must be adjusted individually based on the total number of levels. This way, it can be ensured, that the correct corresponding values are added up during reconstruction. For the example of three layers, this affects the outputs of the highpass filter H_{h1} on level one and the highpass filter H_{h2} of level two. The corresponding delays d_1 and d_2 are also represented in Figure 1.

Calculation of filter bank delays To calculate the overall delay $d_{\text{filterbank}}$ of a filter bank for an SWT all the group delays of the lowpass filters are needed. The group delay of a filter X is denoted by $d_X(\omega)$ depending on the frequency ω . Hence, the group delays of the lowpass decomposition filters are $d_{H_{li}}(\omega)$ with i indicating the corresponding level. Analogous, the delays of the lowpass reconstruction filters are $d_{R_{li}}(\omega)$.

The delay of a filter bank can be calculated as

$$d_{\text{filterbank}}(\omega) = \sum_{i=1}^n d_{H_{li}}(\omega) + \sum_{i=1}^n d_{R_{li}}(\omega). \quad (1)$$

The lowpass filters of the Daubechies wavelets do not have a linear phase. However, convolving the scaling coefficients with the corresponding wavelet coefficients results in symmetric filters with a linear phase response as can be seen in Figure 2 for the db2 wavelet. Thus, the group delay can easily be calculated. For example, with the db2 wavelet, a three-layer filter bank induces a delay of 21 samples. This means, that the SWT and the ISWT delay the signal by about 20.5 ms if a sampling frequency of 1024 Hz is used.

¹ <https://github.com/ime-luebeck/semg-online-ecg-removal>

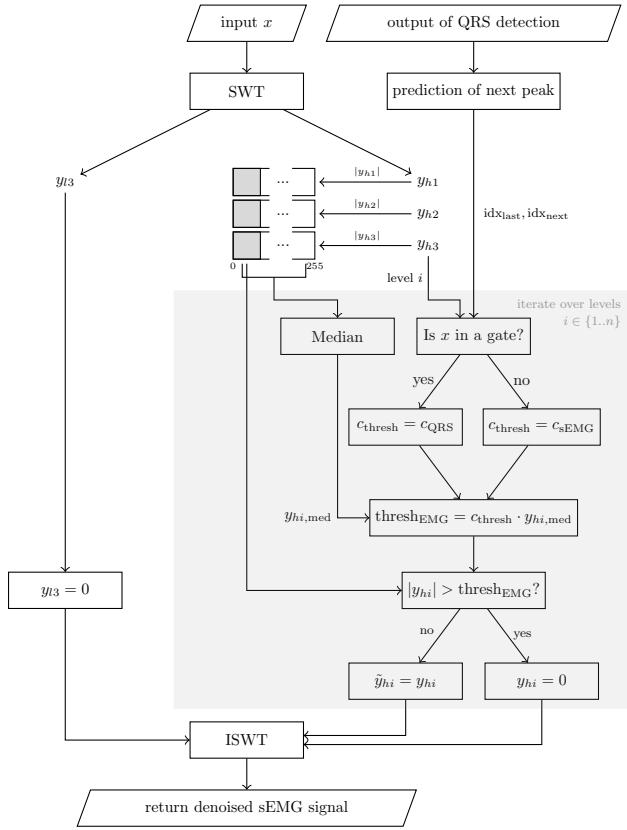


Fig. 3. The flow of the denoising process with input value x . Processes are represented as rectangles and buffers as open rectangles. The spaces in the buffers, where new values are inserted, are displayed in dark gray. The process in the light gray box is executed successively for each detail coefficient y_{hi} with $i \in \{1..n\}$ of the levels of the filter bank.

Prediction of the next R peak As already mentioned, due to the comparatively high delay of the QRS detection output, it cannot be determined online whether a measured sample is part of a QRS complex. To solve this issue, the heart rate is estimated based on previously detected QRS complexes. The algorithm uses the average length of the last four cardiac cycles to predict the position of upcoming QRS complexes. In addition, it also ensures that adaptive thresholding works even if the QRS detection occasionally fails during a measurement.

Adaptive thresholding To enable thresholding dependent on the position in the ECG signal, gates are placed around the QRS complexes. The number of samples n_g within the gates depends on the level i of the wavelet band in the filter bank as

$$n_g = 0.2 \cdot f_s \cdot i, \quad (2)$$

where f_s is the sampling frequency. In order to create those gates, the locations of the peaks in the signal are necessary. To decide whether the last measured value belongs to a QRS region, a simple case distinction can be used. Three possible cases exist for the measured sample to lay within a gate: it has to be considered whether the sample lies within the gate of the last detected peak (case 1) or of the next predicted peak, which may be upcoming (case 2)

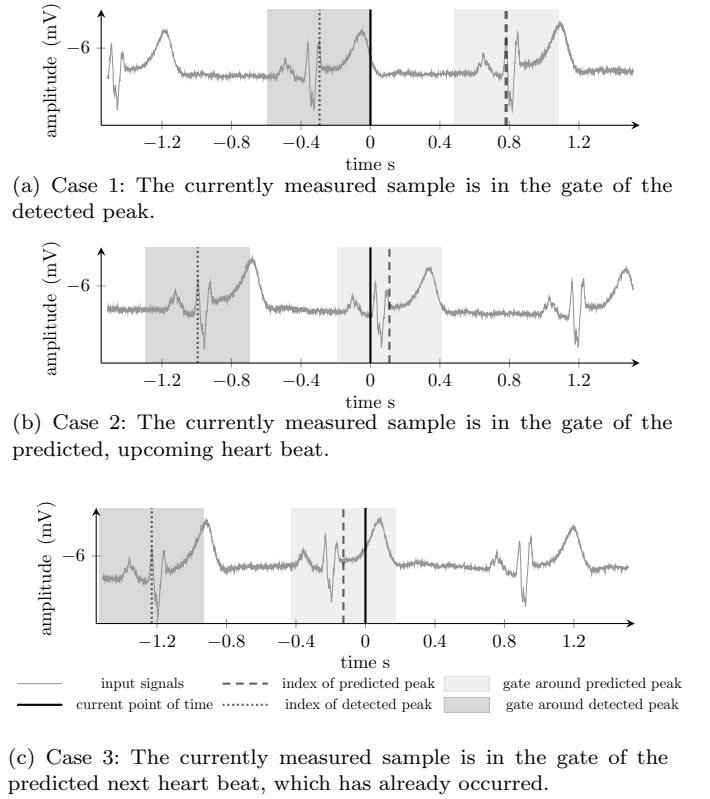


Fig. 4. Examples of the three possible situations, where the currently measured sample is located in a QRS region. Negative time values represent the past and index 0 the current moment.

or lay in the past (case 3). All cases can also be seen in Figure 4.

The hard thresholding process used in the proposed algorithm was developed by Donoho (1995). First, the online algorithm calculates the median of the last 256 absolute coefficients, since it is not possible to perform a centered operation as is commonly done in offline computation. The individual threshold of each value is based on whether the considered value is located in a QRS region. Depending on this, the threshold value $\text{thresh}_{\text{EMG}}$ is calculated using a larger or smaller factor. Values in QRS regions should generally have a lower threshold. In this paper, the factors $c_{\text{QRS}} = 4$ and $c_{\text{sEMG}} = 10$ are used for calculating the thresholds. They were chosen empirically with a similar order of magnitude as in Petersen et al. (2020) and appeared to work well on the data. After setting all detail coefficients above the threshold to zero, the inverse stationary wavelet transform is calculated using the filter bank implementation explained earlier. The resulting value is the denoised sEMG sample.

2.2 Validation

Comparison to the offline algorithm The aspects described in the last chapter differ from the offline algorithm proposed by Petersen et al. (2020). While the offline algorithm creates a mask for the entire signal to assign each value as inside or outside of a gate, in the online algorithm this part is replaced by the described case distinction. It is important to note that this part relies on accurate results

of the peak prediction. In the hard thresholding process of the offline algorithm, a centered median is applied to the absolute of the detail coefficients. In contrast, the online algorithm uses only half the window size for a one-sided median.

Data basis In order to validate the developed algorithm, two sets of signals were considered. The first set, which was also used in the paper of Petersen et al. (2020), contains respiratory sEMG measurements of six subjects with two channels. The second set includes synthetic signals, which each consist of an ECG signal ECG_{clean} and an sEMG signal EMG_{raw} . The ECG is scaled with a factor k_{scale} and added to the sEMG signal forming the synthetic signal

$$EMG_{syn} = EMG_{raw} + k_{scale} \cdot ECG_{clean}. \quad (3)$$

In order to examine the functionality of the algorithm on different SNRs, multiple synthetic signals with varying values for k_{scale} are created. This is done for 20 ECG signals from a public Physionet database from Goldberger et al. (2000), which are the same as used by Petersen et al. (2020). The sEMG signal is an sEMG measurement of the calf muscle, for which we recorded one differential channel using standard measurement equipment as described in Petersen et al. (2020).

Error analysis To validate the general functionality of the online algorithm, two errors are calculated: the normalized removal error norm e_{raw} and the normalized envelope error norm e_{env} , where the envelope of the raw and denoised signal is calculated with a moving mean and a window size of 256 samples. Both errors are introduced in the paper of Petersen et al. (2020) for this purpose and are used on the synthetic signals. This way the filtered sEMG signal $EMG_{cleaned}$ can be compared to the original sEMG signal EMG_{raw} used to create the contaminated signal. The normalized removal and envelope error norms of the result of the online algorithm in comparison to EMG_{raw} are called $e_{raw,online}$ and $e_{env,online}$. Analogous, the errors of the offline algorithm are $e_{raw,offline}$ and $e_{env,offline}$. Comparing the errors of the online algorithm to those of the offline algorithm, the functionality can be assessed. Calculating the difference of the errors of the online algorithm and the offline algorithm leads to

$$\Delta e_{raw} = e_{raw,online} - e_{raw,offline}. \quad (4)$$

With this error, the loss of precision can be estimated. The latter is performed if no true reference is available, which is the case for real respiratory sEMG signals.

SNR-analysis To evaluate the denoising impact of the online algorithm, the signal-to-noise (SNR) improvement attained by the algorithm is calculated. Therefore, the SNRs before and after denoising are computed. In this case, the SNR is considered to be the ratio of sEMG to ECG. The SNR of a respiratory sEMG signal with cardiac artifacts, $SNR\{EMG_{meas}\}$, and of the filtered signal, $SNR\{EMG_{cleaned}\}$, are calculated as in the paper of Petersen et al. (2020).

3. RESULTS

This section presents the results of the validation. Additionally, an excerpt of a respiratory sEMG signal before

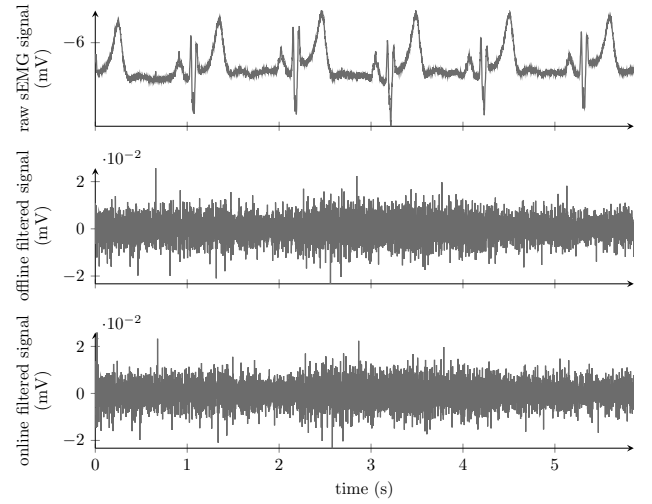


Fig. 5. An extract of a raw respiratory sEMG signal (at the top) and the corresponding offline and online filtered signals (middle and bottom).

and after filtering is presented in Figure 5. It can be seen that no ECG artifacts are visible in the filtered signals and the online and offline filtered signals look very similar.

3.1 Normalized removal and envelope errors

The normalized removal error norm e_{raw} and the normalized envelope error norm e_{env} are based on synthetic signals. These signals are generated by adding an sEMG signal to various ECG signals as described above. In this way 20 signal groups were created with a total of 120 different signals. The generated SNRs range from 0.02 to one. This corresponds to values between approximately -34 dB and 0 dB. For each SNR, the median of all signals was calculated.

Each norm is computed for the online and offline filtered signals and for two highpass filtered signals. To this end, highpass filters with cut-off frequencies of 128 Hz and 200 Hz were chosen. The 128 Hz highpass equals the process of discarding the wavelet band with the lowest frequencies in a three-layer filter bank. The cut-off frequency of 200 Hz is recommended for sEMG ECG denoising by Petersen et al. (2020). These calculations result in the signals $e_{raw,filter}$ and $e_{env,filter}$, where “filter” is either “online”, “offline”, “highpass(128 Hz)” or “highpass(200 Hz)”.

When calculating the two errors, the delay of 21 samples caused by the online SWT is compensated for, resulting in the signal EMG_{online} . All error norms calculated are shown in Figure 6.

In Figure 6a, the normalized removal error norm e_{raw} with the clean sEMG EMG_{raw} as the reference signal is visualized. As can be seen, all errors decrease exponentially. Additionally, the curves for the offline algorithm closely match those for the online algorithm. A similar behaviour can be observed for the envelope errors plotted in Figure 6b. Furthermore, the plots also show, that both highpass filters produce a significantly higher error than the SWT algorithms.

To compare the results of the online and offline algorithms, Figure 6c can be considered. The plot shows an increase

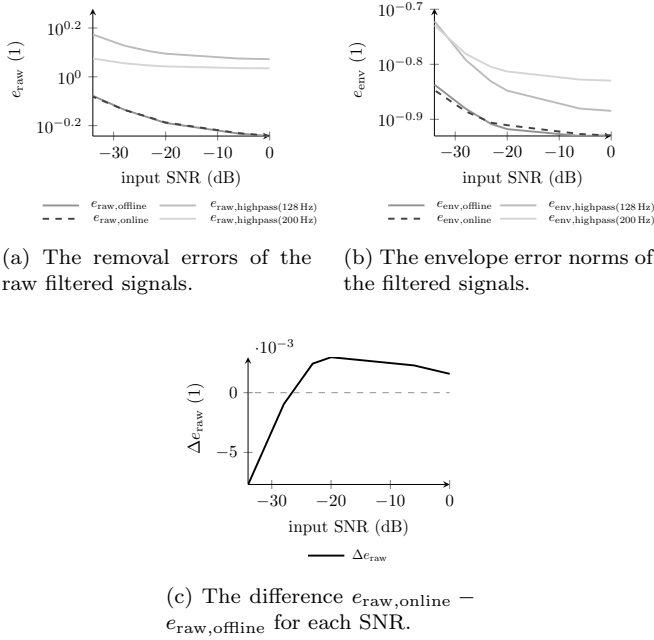


Fig. 6. The medians of all normalized removal and envelope error norms for synthetic signals with different signal-to-noise ratios. All plots show comparisons of online- and offline-algorithm. Plots (a) and (b) additionally show the errors of highpass filters with thresholds of 128 and 200 Hz.

in the curve up to an SNR of about -20 dB, followed by a decrease. For SNRs greater than approximately -26 dB, the error is positive, indicating that the offline algorithm performs better. The errors are between $-8 \cdot 10^{-3}$ and $3 \cdot 10^{-3}$.

3.2 SNR analysis

The online and offline filtered respiratory sEMG signals are evaluated using SNR analysis. To this end, the SNRs of the sEMG measurements of six test subjects are calculated before and after filtering.

Figure 7 shows that the SNRs of the raw measured sEMGs range from approximately -40 dB to -5 dB. In contrast, the SNRs of the filtered signals are in the range of 0 dB to 16 dB. In all cases, the SNRs of the filtered sEMG signals is positive, while the SNRs of the raw measured signals are negative. This means that whereas the ECG dominated the sEMG signal before filtering, the sEMG signal dominates the ECG afterwards. The differences between $\text{SNR}\{\text{EMG}_{\text{online}}\}$ and $\text{SNR}\{\text{EMG}_{\text{offline}}\}$ are generally quite small as the values overlap in most cases. However, it is noticeable that in subject 3 $\text{SNR}\{\text{EMG}_{\text{online}}\}$ is significantly lower than $\text{SNR}\{\text{EMG}_{\text{offline}}\}$. The same applies to channel two of subject 5.

4. DISCUSSION

Before going into detail about the calculated errors, a general overview of potential sources of differences between the results of online and offline algorithm is given.

The prediction of the peaks As the upcoming peaks are only predicted, the actual position of the peaks is not taken

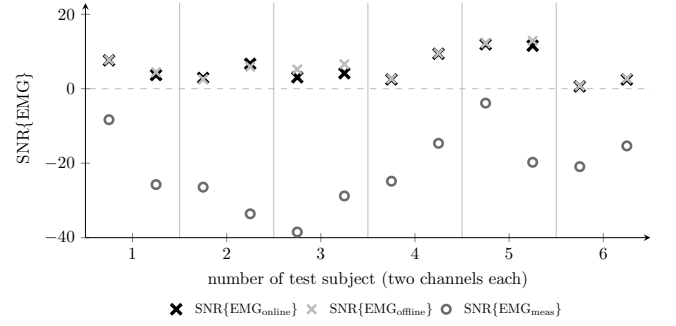


Fig. 7. The SNRs of the online and offline filtered respiratory sEMG signals, $\text{SNR}\{\text{EMG}_{\text{online}}\}$ and $\text{SNR}\{\text{EMG}_{\text{offline}}\}$, in comparison to the SNRs of the original signal, $\text{SNR}\{\text{EMG}_{\text{meas}}\}$, for the two channels of each of the six test subjects.

into account for the gating procedure. Therefore, the gates are not constant in size, as they are in the offline algorithm, and are unlikely to be precisely centered around the peak. Using gates that are too small can result in insufficient ECG removal, while using gates that are too large may lead to excessive loss of sEMG.

Different results in the peak detection If the online algorithm detects a peak in a different location than the offline algorithm, but within the same QRS complex, this may slightly affect the denoising results. However, this does not necessarily mean that the online algorithm will produce worse results than the offline algorithm in this case. While false negatives can be compensated for to some extent, false positives lead to incorrect gates and heart rates, which affects the prediction of upcoming peaks. This can lead to incorrect gating, suppressing more of the sEMG and less of the ECG components.

The one-sided median for the threshold calculation Considering only the samples on one side of a value for a median can give similar results if the signal is very steady or periodic with a small period. However, the more the signal changes over time, the greater the effect of the one-sided median compared to a centred median. In this case, it affects the thresholds for the ECG and therefore the size of the gates.

4.1 Error analysis

As can be seen in Figure 6, the online algorithm works as well as the offline algorithm in most cases, since the calculated errors are very similar. Figure 6c shows that the online algorithm performs slightly better than the offline algorithm at lower SNRs. However, at SNRs over -25 dB the offline algorithm achieves lower errors. As Figures 6a and b show, both algorithms work better at higher SNRs. In cases with very low SNRs, it is important to consider that the ECG used for creating the synthetic signals may contain fragments of other signals, such as EMG signals from other muscles. This can affect the resulting signals of online and offline filtering and lead to higher errors.

Figures 6a and b also show that the proposed algorithm performs significantly better than simple highpass filters. The results of the 128 Hz highpass filter indicate that gating in the wavelet bands with higher frequencies leads

to a more effective ECG suppression, resulting in signals that are much more similar to the original sEMG signal compared to using only the highpass filter.

4.2 SNR analysis

Overall, it can be concluded that the denoising of both algorithms was successful, as the sEMG signal dominates over the ECG signal after filtering. Furthermore, the online algorithm performs as well as the offline algorithm in most cases, since the SNRs of the signals after filtering with the online algorithm are very similar to those of the offline algorithm. However, in subject 3 and channel two of subject 5, the online algorithm produced signals with significantly lower SNRs than the offline algorithm. Possible reasons for this are explained at the beginning of this section. In the case of subject 3, an irregular ECG may have caused a significant difference between the predicted and actual peak positions. For subject 5, movement artifacts may have resulted in larger differences between the one-sided and centred median.

4.3 Delay of the algorithm

The online algorithm introduces a delay of precisely 21 samples when using a wavelet transform with three levels as in our experiments. This is equivalent to a time span of about 20.5 ms between the input of a measured value and the output of the corresponding filtered value. As conventional triggers, such as flow or pressure triggers, have a trigger delay of 98 ± 34 ms (Clement et al., 2011), reducing the delay to 20.5 ms would be a significant improvement. Additionally, this delay would be sufficient for optimizing mechanical ventilation at the bedside.

4.4 Runtime of the algorithm

Table 1: An overview of the runtime of the processes of the online denoising algorithm and the assessed runtime of the whole algorithm per sample. The values were measured for a Python3 implementation on a standard laptop.

Process	Runtime (ms)
Whole denoising algorithm	0.240
QRS detection	0.049
Heart rate calculation	0.006
SWT denoising	0.168

Given that the online algorithm receives inputs at a frequency of 1024 Hz, the processing of each value must be completed in less than $1/1024$ second, which equates to approximately 0.97 ms. The runtime of the algorithm was measured for a Python3 implementation on a standard laptop with an Intel® Core™ i7-1065G7 Processor. As presented in Table 1, the overall denoising algorithm has a runtime of 0.24 ms under these circumstances. This equals approximately one-quarter of the time span between two samples and therefore is sufficient for at least a single-channel measurement.

5. CONCLUSION

In conclusion, the main contribution of this paper is the real-time capability and accessibility of the proposed ex-

tensively tested algorithm for ECG removal from respiratory sEMG signals.

REFERENCES

- Bakkes, T., van Diepen, A., De Bie, A., Montenij, L., Mojoli, F., Bouwman, A., Mischi, M., Woerlee, P., and Turco, S. (2023). Automated detection and classification of patient-ventilator asynchrony by means of machine learning and simulated data. *Computer Methods and Programs in Biomedicine*, 230, 107333.
- Christov, I.I. (2004). Real time electrocardiogram qrs detection using combined adaptive threshold. *BioMedical Engineering OnLine*, 3(1), 28.
- Clement, K.C., Thurman, T.L., Holt, S.J., and Heulitt, M.J. (2011). Neurally triggered breaths reduce trigger delay and improve ventilator response times in ventilated infants with bronchiolitis. *Intensive Care Medicine*, 37(11), 1826–1832.
- Daubechies, I. and Sweldens, W. (1998). Factoring wavelet transforms into lifting steps. *Journal of Fourier Analysis and Applications*, 4(3), 247–269.
- Donoho, D.L. (1995). De-noising by soft-thresholding. *IEEE transactions on information theory*, 41(3), 613–627.
- Goldberger, A.L., Amaral, L.A., Glass, L., Hausdorff, J.M., Ivanov, P.C., Mark, R.G., Mietus, J.E., Moody, G.B., Peng, C.K., and Stanley, H.E. (2000). PhysioBank, physioToolkit, and physionet: components of a new research resource for complex physiologic signals. *circulation*, 101(23), e215–e220.
- Heunks, L. and Ottenheijm, C. (2018). Diaphragm-protective mechanical ventilation to improve outcomes in icu patients? *American Journal of Respiratory and Critical Care Medicine*, 197(2), 150–152.
- Hou, C., Cai, F., Liu, F., Cheng, S., and Wang, H. (2022). A method for removing ecg interference from lumbar emg based on signal segmentation and ssa. *IEEE Sensors Journal*, 22(13), 13309–13317.
- Larsen, R. and Ziegenfuß, T. (2009). *Beatmung: Grundlagen und Praxis*. Springer-Verlag.
- Mauri, T., Cambiaghi, B., Spinelli, E., Langer, T., and Grasselli, G. (2017). Spontaneous breathing: a double-edged sword to handle with care. *Ann Transl Med*, 5(14), 292.
- Pan, J. and Tompkins, W.J. (1985). A real-time qrs detection algorithm. *IEEE Transactions on Biomedical Engineering*, BME-32(3), 230–236.
- Petersen, E., Sauer, J., Grahoff, J., and Rostalski, P. (2020). Removing cardiac artifacts from single-channel respiratory electromyograms. *IEEE Access*, 8, 30905–30917.
- Rafols-de Urquía, M., Estrada, L., Estevez-Piorno, J., Sarlabous, L., Jane, R., and Torres, A. (2018). Evaluation of a wearable device to determine cardiorespiratory parameters from surface diaphragm electromyography. *IEEE J Biomed Health Inform*, 23(5), 1964–1971.
- Sauer, J., Graßhoff, J., Carbon, N.M., Koch, W.M., Weber-Carstens, S., and Rostalski, P. (2024). Automated characterization of patient-ventilator interaction using surface electromyography. *Annals of Intensive Care*, 14(1), 32.